

YOLOv8-BYTE: Ship tracking algorithm using short-time sequence SAR images for disaster response leveraging GeoAI

Muhammad Yasir^a, Liu Shanwei^{a,*}, Xu Mingming^a, Wan Jianhua^a, Sheng Hui^a, Shah Nazir^b, Xin Zhang^c, Arife Tugsan Isiacik Colak^d

^a College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China

^b Department of Computer Science, University of Swabi, Swabi 23430, Pakistan

^c Radar Technology Research Institute, Beijing Institute of Technology, Beijing 100081, China

^d National University of Science and Technology, International Maritime College Oman, Sohar, Oman

ARTICLE INFO

Keywords:

Multi-Object Tracking (MOT)
YOLOv8-BYTE
Tracking Ship
Kalman Filters (KF)
Synthetic Aperture Radar (SAR)
Disaster Response Leveraging GeoAI

ABSTRACT

Ship tracking technology is crucial for emergency rescue in the event of a disaster. Quickly identifying the position and status of vessels is vital for rescue teams to be able to deploy efficiently in disaster areas. When responding to emergencies or natural disasters, ship tracking technology plays a critical role in supporting emergency rescue operations and resource allocation, improving the overall resilience of the maritime transportation system. However, the research on multi-object tracking (MOT) algorithms has primarily focused on optical image datasets. In contrast, image data from synthetic aperture radar (SAR) presents unique challenges, such as defocus interference, a high false alarm rate, and a lack of prior samples. To overcome these particular challenges, we propose a robust MOT algorithm developed for SAR images to achieve effective multi-vessel tracking under difficult imaging conditions. In particular, we optimize the YOLOv8 detection network by introducing a diffusion model-based training method for data augmentation. This method improves the robustness of the network to scaling, rotational and translational deformations. Moreover, an enhanced swin transformer is proposed as a feature extraction network, which strengthens the representation capability of the detection network. Furthermore, the state parameters within the KF technique are enhanced by directly capturing the details of the height and width of the tracking rectangle box. This refinement of the ByteTrack algorithm aims to achieve a more precise and accurate fit of the tracking rectangle to the ship, further improving the overall tracking performance. The experimental results from the ship detection and multiple objects tracking datasets show the impressive performance of the proposed model. With a precision of 97.60%, a recall of 96.36%, and an average precision of 96.72%, the model achieves exceptional detection accuracy with an 18% reduction in model parameters. Furthermore, significant improvements can be observed in key tracking metrics such as HOTA, MOTA and IDF1, with improvements of 4.8%, 8.5% and 6.8% respectively compared to the baseline algorithm, alongside a remarkable 37.5% reduction in IDS. It is noteworthy that the tracker works in real time, achieving an average analysis speed of 47 frames per second. The proposed MOT algorithm achieves state-of-art tracking performance on a SAR image dataset with short time sequences. Therefore, the proposed approach is a compelling solution for ship tracking in SAR imagery.

1. Introduction

Satellite-based synthetic aperture radar (SAR) has gained considerable significance in diverse maritime applications, primarily due to its comprehensive coverage, time-independent data acquisition capabilities, and its ability to operate effectively in all-weather environment (Chen et al., 2023; Chen et al., 2021; Kong et al., 2023; Liu et al., 2023;

Mao et al., 2023; Owda et al., 2023; Yasir et al., 2023a; Zha et al., 2023). The extensive use of SAR in ship detection and monitoring studies highlights its effectiveness (Li et al., 2022a; Yasir et al., 2024; Yasir et al., 2023c). The manual ship localization process in SAR imagery proves to be both difficult and time-consuming, primarily due to the extensive coverage of oceanic areas and the essential expertise required for annotating ship coordinates (Bhattacharjee et al., 2023).

* Corresponding author.

E-mail address: 20110052@upc.edu.cn (L. Shanwei).

<https://doi.org/10.1016/j.jag.2024.103771>

Received 13 February 2024; Received in revised form 29 February 2024; Accepted 16 March 2024

Available online 22 March 2024

1569-8432/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

Deep learning (DL) has garnered considerable attention for automating ship detection and monitoring using SAR images. The effectiveness and reliability of DL models are crucial for the accuracy of ship tracking systems (Zhang et al., 2021a). Developing DL models with robust generalization capabilities across diverse environmental conditions, including variable weather, lighting, and sea states, is imperative (Humayun et al., 2023; Jiang et al., 2019; Liu et al., 2022; Yang et al., 2023). In addition, the maritime domain comprises various vessel types, each with unique characteristics. Consequently, DL models must experience training on versatile datasets to ensure accurate vessel identification and tracking capabilities (Cheng et al., 2023). Minimizing false positives and negatives is paramount, given that identification errors can pose significant risks to maritime security and safety (Cong-An et al., 2022). Adherence to established regulatory standards serves as an additional driver for progress, ensuring the reliability and excellence of ship detection and tracking model. Concurrently, continuous enhancements in deep learning models play a pivotal role in seamlessly integrating these systems with other technologies (Matasci et al., 2021). These efforts drive innovation within the field, effectively address emerging challenges, and enhance the superiority of ship tracking and detection technologies. Various SAR datasets have been employed to develop and evaluate ship detection model (Cheng et al., 2023; Cong-An et al., 2022). While data is a valuable resource for training and evaluating ship detection models, its limited scale could hinder the models' ability to generalize effectively across diverse scenarios (Yasir et al., 2023b).

Distributed object tracking algorithms (DOTA) represent another well-known dataset utilized in ship detection. DOTA poses a challenge to detection algorithms due to its extensive array of object categories (Xia et al., 2018). While these datasets hold value, their scarcity limits the applicability and scalability of models trained with them, potentially introducing complexity when comparing and benchmarking algorithms (Humayun et al., 2023; Matasci et al., 2021). The acquisition of accurately labeled datasets that faithfully reflect real-world scenarios presents a significant challenge, leading to potential biases in the data (Ding et al., 2023; Escorcia-Gutierrez et al., 2022). As a result, ship identification methods may experience reduced reliability when faced with insufficient annotated data for severe weather conditions or specific geographic locations (Lou et al., 2022). Hence, despite the presence of intricate ship detection algorithms in existing databases, their limitations include small scale, lack of standardization, scarcity of specific data, and limited diversity in categories. In recent years, there has been a continual stream of proposed deep learning-based multi-target tracking techniques. (Bewley et al., 2016) introduced the Simple Online and Real-Time Tracking (SORT) algorithm, a two-phase tracking methodology grounded in deep learning objectives. The initial stage involves target detection through a detection model, while the subsequent stage achieves precise target tracking using the Hungarian matching algorithm and matched filtering. However, SORT exclusively relies on Intersection over Union (IOU) as the distance metric for matching, supervising other relevant physical quantities and target category characteristics. Building upon SORT, DeepSORT (Wojke et al., 2017) incorporates motion and appearance information of the object as a novel matching distance metric, significantly enhancing the accuracy of multi-target tracking, especially in dense scenes.

However, the aforementioned methods discard rectangular boxes with low confidence, leading to the loss of tracking for challenging samples, such as occluded targets. Addressing this limitation (Zhang et al., 2022) presented ByteTrack, a novel technique that retains all detected rectangular boxes, even those with low scores. This inclusive strategy significantly enhances the recall rate of the tracking algorithm. Moreover, a succession of single-stage tracking algorithms has been introduced. This category of algorithms integrates the detection and tracking models, delivering end-to-end target tracking results. For instance, (Zhou et al., 2020) introduced CenterTrack, which concurrently offers target information for each frame along with the

corresponding matching relations from the previous frame. (Peng et al., 2020) presented ChainedTracker, a method that processes two consecutive frames as input and simultaneously produces both detection results and target matching outcomes. However, the application of this method to ship target tracking in SAR images faces challenges. Unlike optical images, SAR images feature substantial complex background clutter, significantly impeding target detection performance. Moreover, the high aspect ratio of ship targets poses a challenge when utilizing ByteTrack technology for tracking multiple ships. A potential issue arises where the tracking rectangle may not completely encompass the ship, leading to inaccuracies in the Kalman Filter state vector estimation, particularly the rectangular box width, causing matching errors.

To address this, this article sets improvement goals for both the target detector and tracking algorithm. It first introduces a SAR ship target detection model with enhanced detection performance along with a corresponding training method. Furthermore, an enhancement to the ByteTrack algorithm is proposed, introducing a new Kalman Filter state vector to enhance the tracking efficiency of ship. The contributions of this paper are outlined as follows:

- An innovative approach for efficient object detection, explicitly developed for ship tracking using short-time sequence SAR images within the YOLOv8 model, is presented. First, we applied a diffusion model-based data augmentation approach to enrich the dataset, increase the sample diversity and reduce the risk of overfitting. Then incorporated a swin transformer technique into the feature fusion phase to increase the model's ability to capture long-range dependencies and enhance generalization. Finally, a residual technique with scaled normalization and learnable parameters was integrated into the basic swin transformer model. This improvement allows the algorithm to efficiently learn feature representations of ships, overcoming the challenges of gradient descent optimization.
- During tracking phase, ByteTrack model (Zhang et al., 2022) was used, which is known for its ability to detect low-scoring rectangles. The ByteTrack model was further enhanced by predicting the information of height and width tracking rectangle directly in the Kalman Filter (KF) model.
- The development and evaluation of our proposed approach using short-term SAR imagery from different maritime environments are essential steps towards improving vessel tracking, which is crucial for facilitating efficient emergency rescue operations and strengthening the resilience of maritime transportation systems during disasters.

The remaining sections of the paper are organized: Section 2 presents the material and methods. Section 3 delineates the results and discussion of the proposed model. Section 4 summarizes the conclusion and outlines avenues for future work.

2. Material and methods

We propose a framework for SAR ship tracking, called YOLOv8-BYTE. It is consisting of a ship detection module and a tracking module. Specifically, the ship detection module is designed to obtain the positions of ships in a single-frame image. An improved YOLOv8 is presented to increase the representational capacity of the detection network for SAR ship targets (Section. 2.3). The tracking module is used to match detected ship targets across multiple frames, and an enhanced tracking model based on ByteTrack is proposed to enhance the multi-object matching accuracy of the tracking module (Section 2.6). In the following subsections, information about the utilized datasets is presented, and the characteristics of the detection and tracking modules and the description of evaluation metrics are provided.

Table 1

List of the utilized datasets.

Satellite	Location	Imaging Time	Time Difference	Resolution	Polarization Mode
RADARSAT-2	Malacca Strait	1/7/2012 22:51:06	152 Second	2.50*2.50 m	VV
TerraSAR-X	Malacca Strait	1/7/2012 22:53:38		2.00*2.00 m	VV
TerraSAR-X	ZheJiang	12/12/2018 09:19:24	41 Minute 36 s	2.08*2.08 m	HH
RADARSAT-2	ZheJiang	12/12/2018 10:01:00		1.95*1.95 m	HH

2.1. Dataset

This research obtained two sets of satellite-based SAR images, including Radarsat-2 and TerraSAR-X for the Strait of Malacca and Zhejiang. First, Radarsat-2 imagery was acquired in the VV polarization mode over the Malacca Strait on January 7, 2012, at 22:51:06. Nearly concurrently, TerraSAR-X also provided imagery in the VV polarization mode, with a slightly higher resolution of 2.00 x 2.00 m. Second, in the Zhejiang region, TerraSAR-X was acquired on December 12, 2018, at 09:19:24 in the HH polarization mode, while RADARSAT-2 was acquired in the same HH polarization at 10:01:00. Table 1 lists the characteristics of the utilized datasets.

2.2. Diffusion model-based data augmentation

The model's translation, scale, and rotation invariance play a pivotal role in the detection of multi-scale and multi-angle ship. Data augmentation techniques like Copy-Paste (Ghiasi et al., 2021) enhance the model's scale invariance by applying scaling transformations to the targets. However, these methods introduce image distortion, creating a significant disparity between augmented and real images. Moreover, researchers enhance the model's rotation invariance by rotating the original image; nevertheless, this approach does not introduce new targets or backgrounds to the dataset.

Ship-Go (Zhang et al., 2024) is an instance-to-image object detection

dataset augmentation method based on the diffusion model. This technique can integrate any set of target instances into SAR images with specified background types. It facilitates the introduction of new targets into the dataset and, but it also enables the generation of novel backgrounds, resulting in more realistic images. Throughout the generation process, the model utilizes target instance variables C_{VI} and background type variables C_{EP} as conditions. A multi-step denoising approach progressively transforms pixels in the noise area from a Gaussian distribution into a sample distribution of clutter scattering intensity under the specified background type. Eventually, this process yields a more authentic SAR image. The denoising process is executed by a deep neural network f_θ , implying that the network's training essentially involves refining the denoising capability, enabling accurate estimation of the noise ϵ_θ as follows Eq. (1):

$$L = E_{(x_t, C_{EP}, C_{VI})} E_{(t, \epsilon_t)} [\|\epsilon_t - f_\theta(x_t, C_{EP}, C_{VI})\|^2] \quad (1)$$

where x_t represents the sample to be denoised in the t-th denoising step. However, in the Ship-Go approach, the author mainly focused on the improvement of background diversity, with less emphasis on showcasing its full potential in terms of target diversity. Drawing inspiration from Ship-Go, we have enhanced this data augmentation method. Our improved approach ensures background diversity and considers target diversity in terms of rotation, translation, and, scale aspects.

Specifically, the instance segmentation algorithm is initially employed to segment ship targets in the dataset, obtain instance masks, and construct a ship instance database. Before feeding the target instances into Ship-Go, we randomly apply various geometric transformations to the target, including rotation and scaling. For rotation operations, pre-defined 8 sets of rotation directions: $[0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ]$. Regarding scaling operations, 3 sets of scales are preset: $[0.5, 1, 2]$. A random rotation angle and scaling factor are selected during the geometric transformation of a target instance. Furthermore, a set of coordinates $[x_i, x_y]$ is randomly obtained for each transformed instance o_i , and the instance is placed in the corresponding position of the image. Note that, we prevent multi-target overlaps by checking for intersections between instance masks.

Similarly, applying the same transformation to the annotations of the original bounding boxes allows obtaining instance-level annotations for the generated images. Finally, the detection model is trained using the generated images and the corresponding annotations. The proposed data

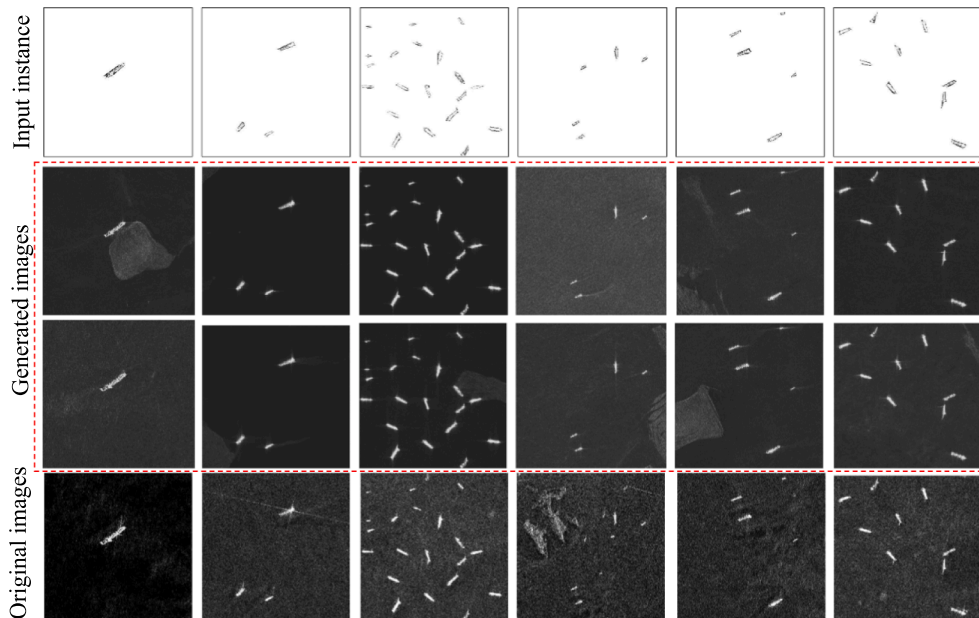


Fig. 1. Illustrated samples of the data augmentation process.

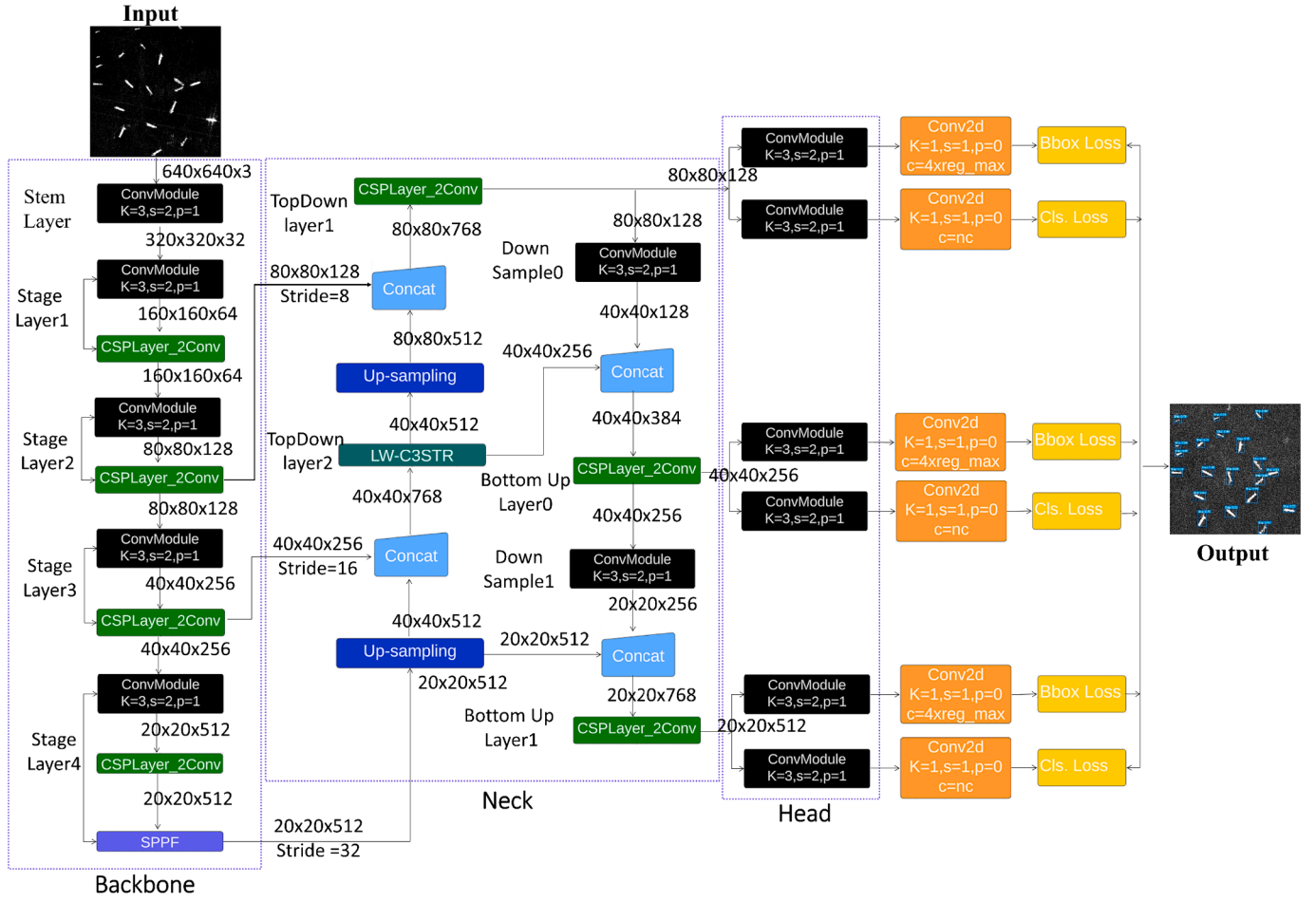


Fig. 2. Overview of the proposed model.

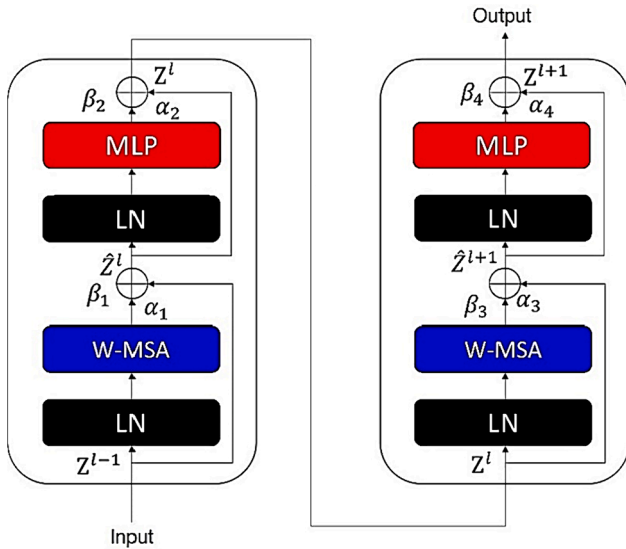


Fig. 3. The enhanced Swin Transformer Block's structure.

augmentation method not only ensures an increase in background diversity but also enhances the model's target diversity in terms of scale, rotation, and translation. Fig. 1 illustrates some instances from the dataset.

2.3. YOLOv8 model design for ship detection

The YOLO model has significantly influenced the field of computer vision, prompting researchers to continuously improve and expand its capabilities. YOLOv8, developed by Ultralytics and released on January 10, 2023, is the latest evolution of this series of algorithms. Compared to its predecessors, including such notable models as YOLOv5 and YOLOv7, YOLOv8 sets new standards with its superior recognition accuracy and faster processing speed, making it a breakthrough solution for various computer vision applications. In the current study, we developed a detection model by integrating the LW-swin transformer deep learning algorithm with YOLOv8s. To enhance the model's performance and robustness, we employed two key strategies. Firstly, we applied data augmentation using a generative model, thereby augmenting the quantity and diversity of samples. Secondly, added the swin transformer network, which significantly raised the model's overall performance by enhancing its capacity for generalization and addressing long-distance dependencies in models. Ultimately, enhanced the swin transformer further by incorporating a residual network with scaled normalization and learnable parameters, facilitating more effective feature fusion. The network architecture is visually illustrated in Fig. 2.

2.4. Enhance swin transformer

A unique transformer architecture known as swin transformer (Liu et al., 2021). It achieves SOTA performance on several computer vision tasks by effectively processing huge amounts of image and video data by utilizing a hierarchical attention system. The swin transformer uses a hierarchical attention system that contrasts with the flat attention

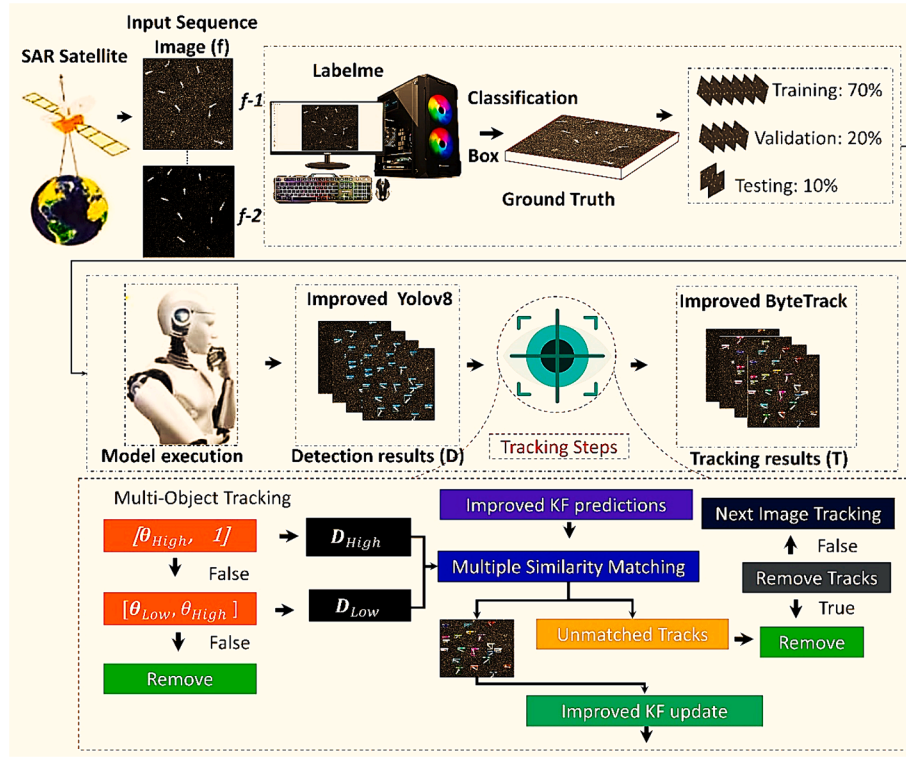


Fig. 4. Flow design of multi-object tracking (MOT) steps for ship tracking using short-time sequence SAR images.

mechanism of traditional transformers. While traditional transformers consider both local and global information in each level of attention, swin transformer's hierarchical design provides a more structured approach that improves the ability to capture different dependencies at different levels. This approach reduces the complexity of the calculations, which becomes a significant problem as the size of the input data increases, by calculating attention at multiple levels independently. After splitting the input image into blocks, the swin transformer handles each block as a separate tiny image. Information processing at the block level uses a layered transformer structure, with a tiny transformer within each layer. A global fusion of information is the outcome of information transmitted across layers. Fig. 3 illustrates the structure of the enhanced Swin Transformer Block.

Furthermore, the swin transformer incorporates cross-tier connectivity to facilitate the exchange of information between different tiers, thereby augmenting overall performance. In addition, swin transformer employs various strategies to enhance model efficiency, complementing its hierarchical design concept. For example, the swin transformer employs a local window-based technique to reduce the computational cost associated with position encoding. This research contributes to the field by enhancing the swin transformer by incorporating scaled normalization and residual connectivity with learnable parameters. This approach enables adaptive learning of feature correlations by leveraging swin transformer's and residuals' strength. Deep learning approaches address the impact of duplicate features by determining the weighting of outcomes for data streams with distinct paths. The two learning parameters start with an initial value of one and are later adjusted by normalization before the weighted summation is performed. This scaling technique involves dividing each value by the total, enabling the data to be scaled to the $[0,1]$ array. This sustains the relative proportionality of the base data and facilitates easier calculations. Moreover, this approach exhibits insensitivity to outlier values, ensuring that the outcomes remain minimally affected even if there are abnormalities in the data. Normalization in neural networks offers various advantages, including enhanced accuracy, accelerated training, and increased model stability

and generalization. The normalization Eq. (2) is as follows:

$$W_i = \frac{x_i}{\sum_{j=0}^n x_j} \quad i \in [0, 1], j \in [0, 1] \quad (2)$$

where:

- xi: un-normalized learning parameters.
- wi: Normalized parameters.

2.5. Metric evaluation metrics for object detection

In this study, four metrics were used to evaluate ship detection, namely recall (R), average precision (AP), and precision (P), demarcated as in Eq. (3), and the count of algorithm parameters (params). The criteria for correct detection of vessels in all tests were based on an intersection over union (IoU) ≥ 0.5 .

$$\begin{cases} P = \frac{TP}{TP + FP} \\ R = \frac{TP}{TP + FN} \\ AP = \int_0^1 PRdR \end{cases} \quad (3)$$

2.6. Ship multi object tracking

In this study, the ByteTrack technique was refined to facilitate the multi object tracking of ships, as illustrated in Fig. 4. The detection outcomes of the improved YOLOv8 algorithm are categorized into low and high-score detection rectangles using two confidence intervals $[\theta_{low}, \theta_{High}]$ and $[\theta_{High}, 1]$. Various matching techniques are applied: first, trajectories T are matched with a high-value detection rectangle, D_{High} un-matched high-value detection rectangle creates new tracks T_{New} , and the becoming tracks D_{remian} are matched with low value detection rectangle D_{low} . Track collection T is modified utilized the Kalman filter.

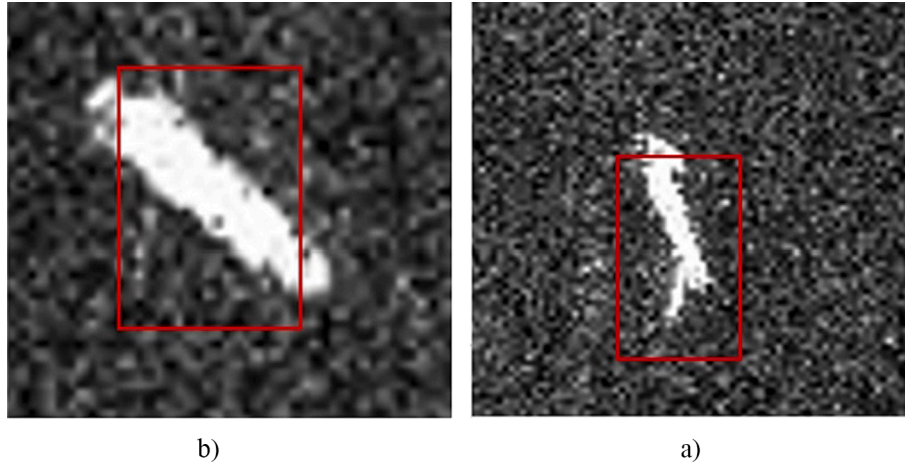


Fig. 5. Challenges in the ByteTrack algorithm for ships tracking. (a) illustration the issue where the detection box fails to adequately surround the ship, (b) depicting a scenario where the detection box includes additional background information.

Table 2

List the comparative analysis of ship detection results using diverse models.

Algorithm	R (%)	P (%)	AP (%)	Params
YOLOv7	94.61	91.80	90.10	4.76×10^8
YOLOv8	94.25	91.21	92.35	3.74×10^8
YOLOv8-x	94.29	92.26	93.45	7.66×10^8
Our Model	96.36	97.60	96.72	3.00×10^8

This method exploits the correlation between the detection rectangle and existing tracks to discard background noise from low-scoring detections, ensuring the retention of high-confidence detections. This approach effectively separates low and high-scoring detections by relying on the similarity between the identify rectangle and established tracks, eliminating false positives while preserving accurate results. This approach effectively unveils the true object, even in scenarios involving blurriness, multiple targets within a scene, and varying sizes, thereby reducing instances of unsuccessful vessel detections and enhancing the continuity of vessel tracking. In the multi object tracking experiment for ships, the threshold value for detecting rectangles θ_{High} with a high hit rate is set at 0.6, and that for detecting rectangles θ_{low} with a low hit rate at 0.3.

2.6.1. Enhanced tracking model for ship

In this section, the enhancements to the ByteTrack algorithm are delineated. Section 3.3.1 is dedicated to the discussion of the experiments conducted and the explication of specific outcomes. The KF technique is generally utilized to model target movements (Tan et al., 2022; Zheng et al., 2023). It predicts and upgrades target trajectories utilizing a linear observation model and a constant velocity (Bewley et al., 2016). By applying the ByteTrack method, the KF creates a step-by-step prediction of the current frame using the previous frame's estimate. By applying the ByteTrack technique, the KF creates a step-by-step prediction of the present frame by using the estimated preceding frame. Eq. (4) illustrates the procedure for calculating the ship's location and the covariance matrix in this research utilizing the detected target position from the YOLOv8 network.

Table 3

Ablation test.

No	+Swin Transformer	+Learnable residuals	+Data enhancement	R (%)	P (%)	AP (%)	Params
1	×	×	×	91.80	92.45	90.90	3.76×10^8
2	✓	×	×	92.21	92.40	90.14	3.07×10^8
3	✓	✓	×	92.12	92.62	93.56	7.68×10^8
4	✓	✓	✓	96.36	97.60	96.72	3.00×10^8

$$\begin{cases} \hat{x}_{t|t-1} = F_t \hat{x}_{t-1|t-1} + B_t u_{t-1} \\ P_{t|t-1} = F_t P_{t-1|t-1} F_t^T + Q \\ K_t = P_{t|t-1} H^T (H P_{t|t-1} H^T + R)^{-1} \\ \hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t (Z_t - H \hat{x}_{t|t-1}) \\ P_{t|t} = P_{t|t-1} - K_t H P_{t|t-1} \end{cases} \quad (4)$$

The symbol, $\hat{x}_{t|t-1}$ represents the prior state estimate for a given frame t , obtained from the prediction formula, where B_t , F_t signifies the measurement matrix and state transition. $\hat{x}_{t-1|t-1}$ is the posterior state estimate of frame $t-1$. u_{t-1} signifies the disturbance of frame $t-1$, $P_{t|t-1}$, and $P_{t-1|t-1}$ are the prior estimate covariance and posterior estimate covariance of frame t respectively. Q denotes the covariance of the system disturbance, reflecting the overall system stability. The adjustment stage, KF technique refines the value predicted in the previous phase by incorporating the observed value from the current frame. This refinement process increases the accuracy of the estimation.

The adjustment of the tracking position and the covariance matrix is determined by comparing the observed and the predicted ship position, as described in Eq (5). In this equation, k_t stands for the gain matrix of the KF, while H denotes the transformation matrix from the state variable to the observation variable. This matrix defining the connection between the state variable and the observed variables. The measurement disturbance matrix is denoted by R , $\hat{x}_{t|t}$ stands for the posterior state estimation, z_t corresponds to the measured value and $p_{t|t}$ denotes the posterior estimation covariance for the frame t . ByteTrack technique utilizes the KF state vector for both the associated and the predicted tracking rectangle. However, since the KF assumes a constant velocity model, its state estimation can lead to suboptimal rectangle box shapes in contrast to the rectangle box generated by object detector. As shown in Fig. 5, the problem may occur that the tracking rectangle does not fit completely and adequately on the ship when the ByteTrack technique is used to track multiple ships. ByteTrack technique utilize of the KF state vector to determine the tracking rectangle aspect ratio leads to an imprecise determination of the tracking rectangle width. To overcome

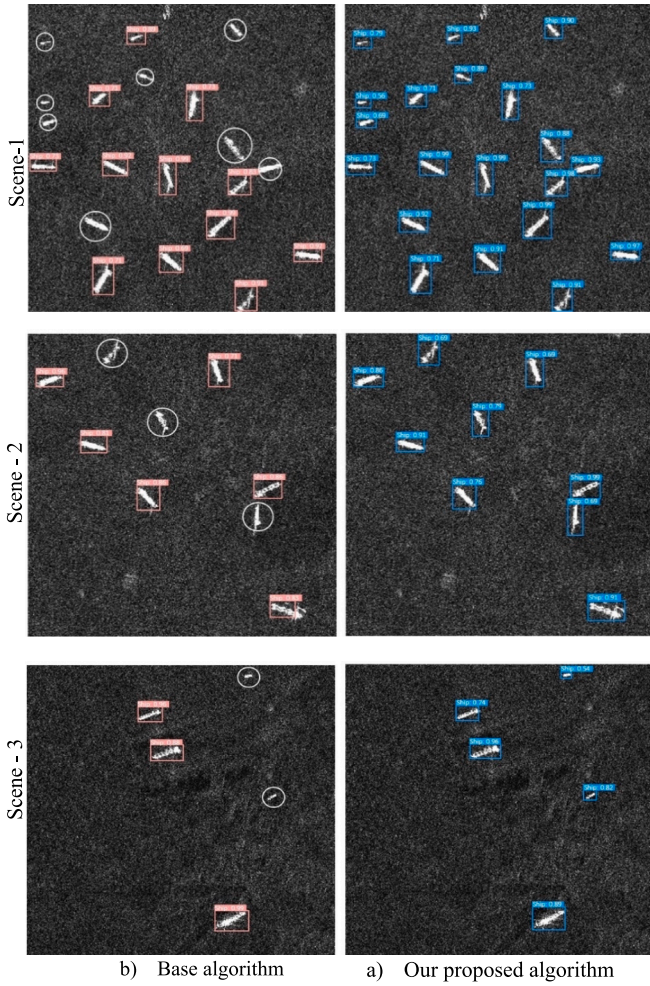


Fig. 6. Comparison of ship instance detection between the baseline model and the proposed model, a) baseline and, b) proposed model detection results. Noted that the yellow color shows false detection while the green color shows the miss detection.

Table 4

Outcomes of the performance evaluation of a ship tracking dataset.

Index	MOTA (%)	HOTA (%)	MOTP (%)	IDF1 (%)	IDS	FPS/(f/s)
1	94.9	87.6	88.6	97.7	1	47.4
2	83.7	68.7	83.9	85.6	7	46.8
3	69.5	69.9	70.5	84.2	3	46.7
4	79.6	71.5	80.1	89.8	0	47.1
Overall	76.3	68.8	83.2	78.9	85	47.0

Table 5

Comparison results of the tracker between the base model and proposed model.

Model	MOTA (%)	HOTA (%)	MOTP (%)	IDF1 (%)	IDS	FPS/(f/s)
YOLOv8 + ByteTrack (base)	68.3	64.2	81.3	73.0	136	56.8
YOLOv8-BYTE (Our model)	76.8	69.0	84.0	79.8	85	47.0

this challenge, our paper proposes a change to the state vector within the ByteTrack technique. The revised state vector, expressed as $x = (x_c, y_c, w, h, \dot{x}_c, \dot{y}_c, \dot{w}, \dot{h})$, includes various variables. Here, (x_c, y_c) stands for the coordinates of the center position of the ship, h, w for the height and width of the tracking rectangle box. In addition, $(\dot{x}_c, \dot{y}_c, \dot{w}, \dot{h})$ denotes the speed of the associated variables in the image coordinates for (x_c, y_c, h, w) .

This modification aims to enhance the accuracy of tracking boxes, specifically tailored for ship tracking scenarios. The Kalman Filter algorithm now directly predicts and updates the height and width of the tracking box. After adjusting the state variables, the state and estimation vector x_t and z_t , state covariance matrix Q , and estimation disturbance matrix R in the KF technique were revised accordingly in Eq. (5).

$$\begin{cases} x_t = [x_c(t), y_c(t), w(t), h(t), \dot{x}_c(t), \dot{y}_c(t), \dot{w}(t), \dot{h}(t)]^T \\ z_t = [\dot{x}_c(t), \dot{y}_c(t), \dot{w}(t), \dot{h}(t)]^T \\ Q = \text{diag}((\delta_p \hat{w}_{t-1|t-1})^2, (\delta_v \hat{h}_{t-1|t-1})^2, (\delta_p \hat{w}_{t-1|t-1})^2, (\delta_p \hat{h}_{t-1|t-1})^2, (\delta_p \hat{w}_{t-1|t-1})^2, (\delta_v \hat{h}_{t-1|t-1})^2, (\delta_p \hat{w}_{t-1|t-1})^2, (\delta_v \hat{h}_{t-1|t-1})^2) \\ R = \text{diag}((\delta_m \hat{w}_{t-1|t-1})^2, (\delta_m \hat{h}_{t-1|t-1})^2, (\delta_m \hat{w}_{t-1|t-1})^2, (\delta_m \hat{h}_{t-1|t-1})^2) \end{cases} \quad (5)$$

In this context, $h(t), w(t)$ signifies the height and width of the tracking rectangle, and $[x_c(t), y_c(t)]$ denotes the center position coordinates of the object. $[\dot{x}_c(t), \dot{y}_c(t), \dot{w}(t), \dot{h}(t)]$ represents the velocity of associated variable coordinates in the image for $[x_c(t), y_c(t), w(t)]$. Additionally, $\hat{w}_{t-1|t-1}$ and $\hat{h}_{t-1|t-1}$ are the posterior state estimates for height and width of the tracking rectangle in the frame $t-1$, while $\hat{w}_{t|t-1}$ and $\hat{h}_{t|t-1}$ are the prior approximations for the height and width of the tracking rectangle in the frame $t-1$. Following the adjustment of the state variable, the ByteTrack model exhibits enhanced alignment with tracking boxes for ship detection, facilitated by disturbance factors denoted as $\delta_p, \delta_v, \delta_m$.

2.6.2. Evaluation metric for MOT tracker

In this study, the ship tracking approach was assessed using various metrics, including: Identification F1 (IDF1), Multi-Object Tracking Precision (MOTP), High Order Tracking Accuracy (HOTA) (Wang et al., 2022), and Multi-Object Tracking Accuracy (MOTA) (Wang et al., 2022). First, IDF1 measures how well the tracker performs. Greater the value indicates that the model can accurately track an object for a longer time (Li et al., 2022b). The localization accuracy of a detector was determined using MOTP metric. Moreover, HOTA introduces a higher-dimensional index to assess tracking accuracy, offering a more comprehensive and well-balanced measure of multi-object tracker performance.

Eq. (6) illustrates the formula for calculating HOTA, where DetA stands for recognition accuracy and AssA for association accuracy. FP the count of negative samples predicted by the algorithm to be positive, TP denotes the count of true identified actual ships, C is a position within TP that allows the identification of a unique ground truth (GT) trajectory, and FN is the count of false alarms. According to Eq. (7), A(c) signifies the association accuracy, TPA(c) denotes the accuracy of true association, while FNA(c) signifies the accuracy of false association prediction. Furthermore, FPA(c) quantifies the accuracy of predicted trajectories that are incorrect or without association.

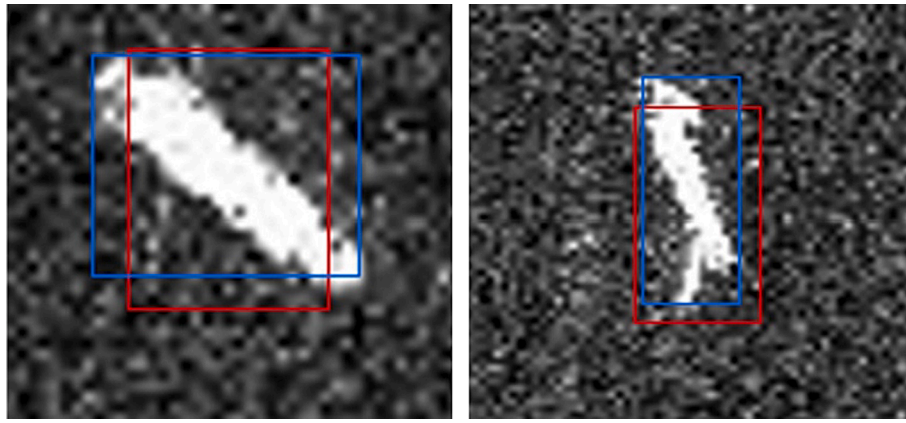


Fig. 7. Comparison of tracking results between the baseline model and improved model by the Kalman Filter technique. The red boxes represent the tracking results of the baseline model, while the blue box indicate the improved tracking detection result of the ByteTrack technique for ship tracking.

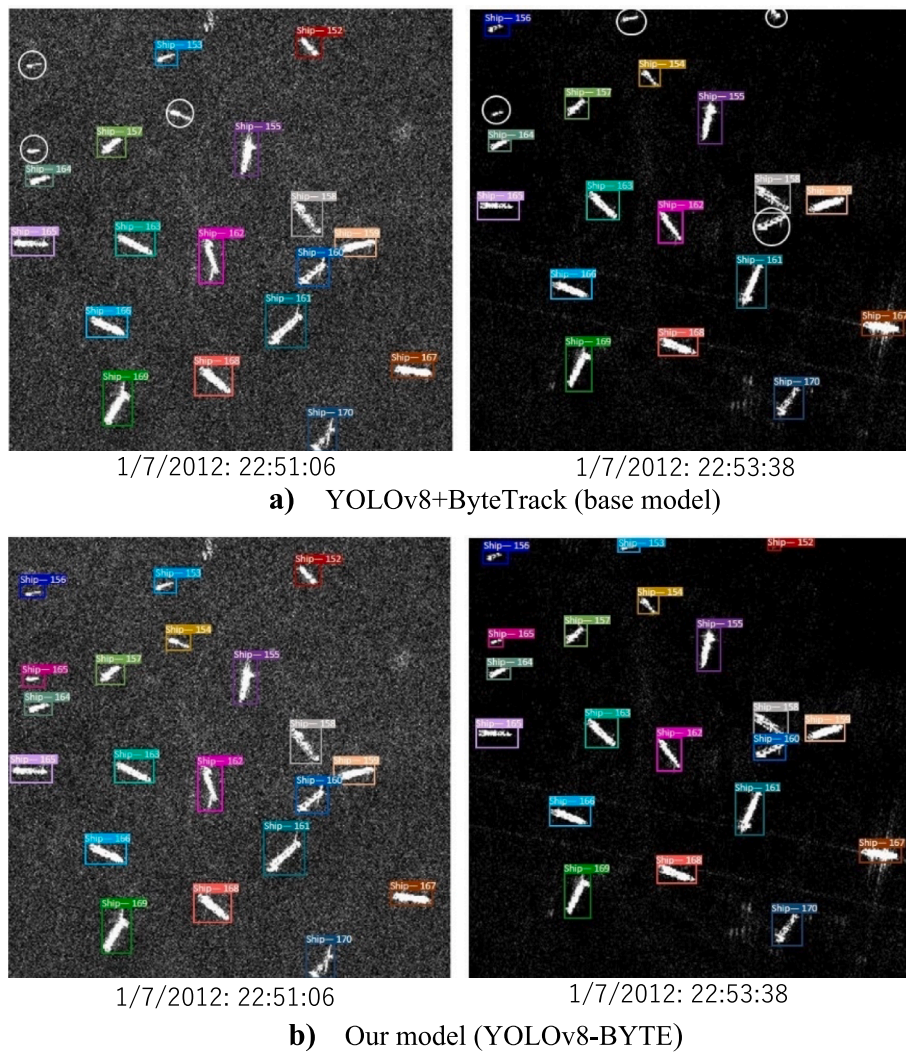


Fig. 8. Comparison between the proposed and baseline trackers algorithms. The white circle indicates a ship that was not successfully tracked. (a) shows the results of baseline model tracker results, while (b) illustrates our proposed YOLOv8-BYTE tracker results.

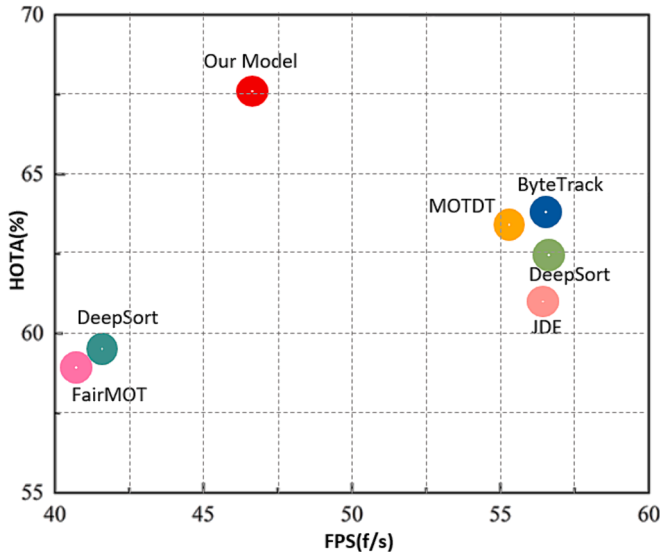


Fig. 9. Comprehensive performance comparison of different trackers techniques for ship using short time sequence SAR images.

$$\left\{ \begin{array}{l}
 HOTA = \sqrt{DetA.AssA} = \sqrt{\frac{\sum_{c \in TP} A(c)}{TP + FN + FP}} \\
 A(c) = \frac{TPA(c)}{TPA(c) + FPA(c) + FNA(c)} \\
 MOTA = 1 - \frac{\sum_i FP + FN + IDS}{\sum_i g_i} \\
 MOTP = \frac{\sum_i d_{i,t}}{\sum_i c_i} \times 100\% \\
 IDF1 = \frac{2IDTP}{2IDTP + IDFP + IDFN} \times 100\%
 \end{array} \right. \quad (6)$$

The MOTA, MOTP, and IDF1 calculation is expressed by Eq. (6), where FN indicates the total count of missed recognitions in a frame t , while the total count of false positives in a frame t is indicated by FP. IDS signifies the count of object marker ID changes throughout tracking in a frame t . g_t stands for the count of object detected in real time t . $d_{i,t}$ signifies the distance between the given and its corresponding hypothetical position in frame t , i signifies the current detected target, and c_t signifies the count of matches between object and hypothesis positions in frame t . IDF1 has three components: IDTP is the overall count of truly tracking targets with unmodified ID, IDFP is the overall count of false tracking targets with unmodified ID, and IDFN is the overall count of object misplaced in tracking with unmodified ID. This study evaluates algorithm efficiency using two additional metrics: the average FPS and the count of detected switches (IDS). A greater score for MOTP, MOTA, IDF1, and FPS, along with a smaller value for IDS, indicates better algorithm efficiency.

3. Results and discussions

3.1. Configuration settings

Training for all of the above models was conducted on Windows 10 Professional Edition, using a system environment with 16 GB of RAM and an Intel(R) Core (TM) i7-12700KF@3.61 GHz CPU. The graphics card used in this study was the NVIDIA GeForce RTX 3080 Ti.

3.2. Analysis and accuracy evaluation of ship detection results

3.2.1. Comparative analysis of the different algorithms

The YOLOv7 and YOLOv8 models are trained with the dataset of this study and their performance is determined utilizing the test dataset. The outcomes are illustrated in Table 2 and are intended to validate the recognition performance of the proposed algorithm in this paper. From Table 2, the R, P and AP values of our model compared to YOLOv7 and the original YOLOv8 are 1.75 %, 5.8 % and 6.62 % better than the original YOLOv8 and 2.07 %, 6.39 % and 4.37 % better than YOLOv7, respectively. R and AP are increased compared with YOLOv8-x. Therefore, the P-value is a bit smaller. Therefore, our algorithm in this work has the fewest parameters among the four algorithms: 1.76×10^8 and 0.74×10^8 less than YOLOv7 and YOLOv8, respectively, and only 40 % of YOLOv8-x.

3.2.2. Ablation study for model enhancement

The ablation study was carried out to test further the efficiency of the optimization algorithm utilized in the current work and evaluate the enhanced efficiency of the merged ship identification algorithm. The model was trained using the original Swin Transformer, learnable residuals and data enhancement. The detection efficiency of the network was analyzed in dissimilar scenarios, and Table 3 illustrates the corresponding results.

As shown in Table 3, ablation studies are critical to model development, especially in deep learning for image processing. They provide invaluable insights into how each component of a model contributes to its overall performance. In the context of the YOLOv8-BYTE algorithm, which is designed explicitly for short-time sequence SAR images, understanding these components is crucial for refining the algorithm's efficiency and accuracy. Table 3 lists the outcomes of the ablation experiment, where we systematically explored the impact of selectively removing or modifying specific components within our model.

The experiment considered three main components: swin transformer, learnable residuals and data enhancement. These components were either included ($\sqrt{}$) or excluded ($\times$) in different configurations to assess their individual and combined effects on the model's performance. The performance metrics include recall, precision, average precision, and the number of parameters (Params).

- **Configuration 1:** This baseline model configuration, lacking both swin transformer, learnable residuals and data enhancement, achieved a recall of 91.80 %, a precision of 92.45 %, and an AP of 90.90 % with 3.76×10^8 parameters. This setup serves as a reference point for understanding the contributions of swin transformer, learnable residuals and data enhancement.
- **Configuration 2:** The inclusion of swin transformer while excluding learnable residuals and data enhancement resulted in a slight increase in recall (92.21 %) but a marginal decrease in AP (90.14 %). Precision remained relatively stable (92.40 %). Notably, this configuration reduced the model's parameters to 3.07×10^8 , indicating that swin transformer contributes to the model's efficiency.
- **Configuration 3:** Incorporating swin transformer and learnable residuals led to a significant improvement in AP (93.56 %), with recall and precision also showing improvements of 92.12 % and 92.62 %, respectively. However, this configuration required a substantial parameter increase (7.68×10^8), suggesting that while swin transformer and learnable residuals enhances accuracy, it does so at the cost of increased computational complexity.
- **Configuration 4:** The final configuration, which includes swin transformer, learnable residuals and data enhancement, yielded the most impressive results. The model attained a recall of 96.36 %, a Precision of 97.60 %, and an AP of 96.72 %, all while maintaining a relatively low parameter count of 3.00×10^8 . This configuration improved accuracy significantly and managed to do so efficiently,

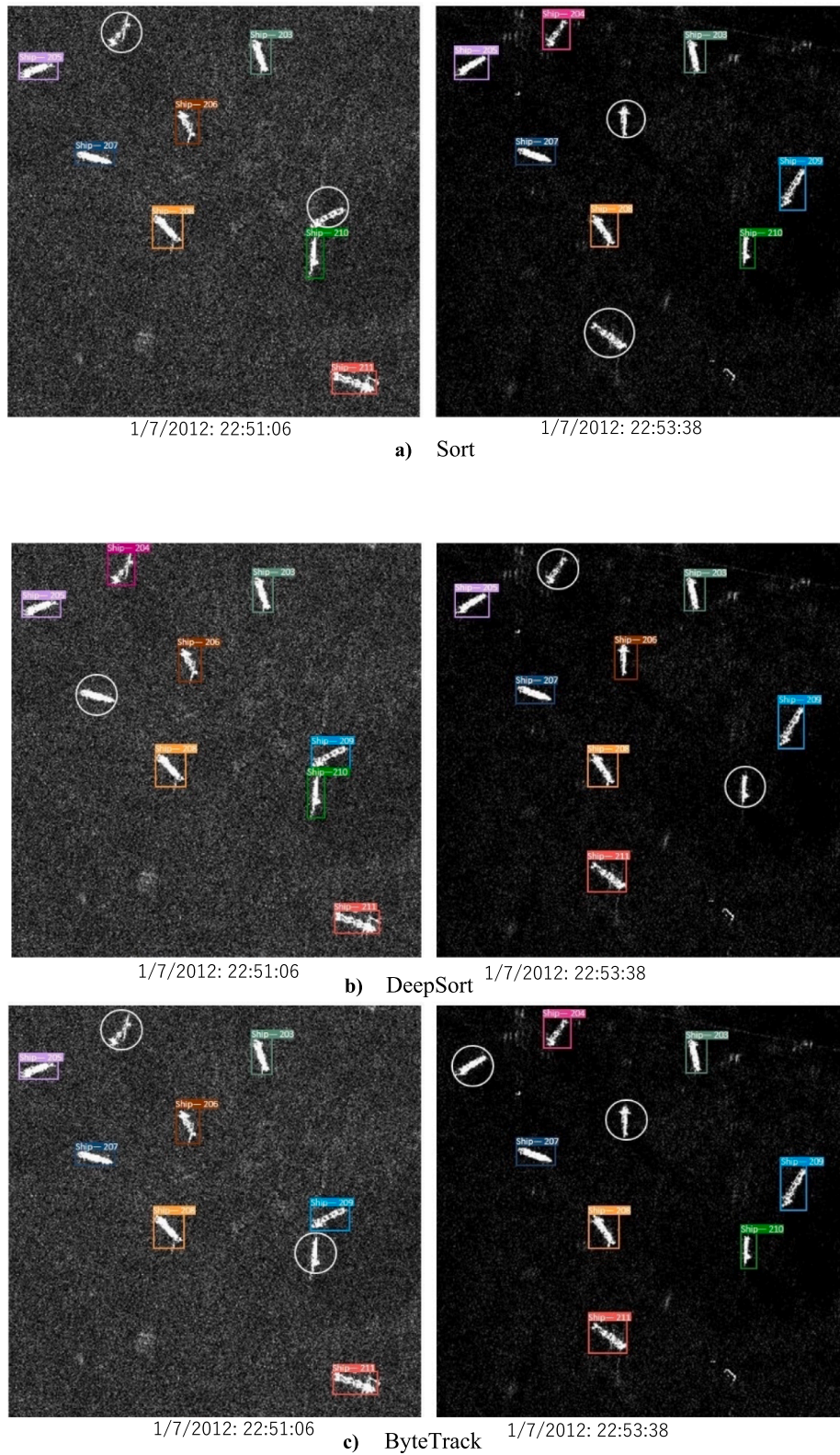


Fig. 10. Comparison between the original and the proposed trackers model. The white circle indicates a target that was not successfully tracked. (a) the tracking results of YOLOv8 + Sort. (b) the tracking results of YOLOv8 + DeepSort. (c) the tracking results of YOLOv8 + ByteTrack. (d) the results of our proposed algorithm for ship tracking.

highlighting the synergistic effect of combining swin transformer, learnable residuals and data enhancement.

In summary, the ablation study described in Table 3 demonstrates the significant effect of both swin transformer, learnable residuals and data enhancement on improving the efficiency of the YOLOv8-BYTE

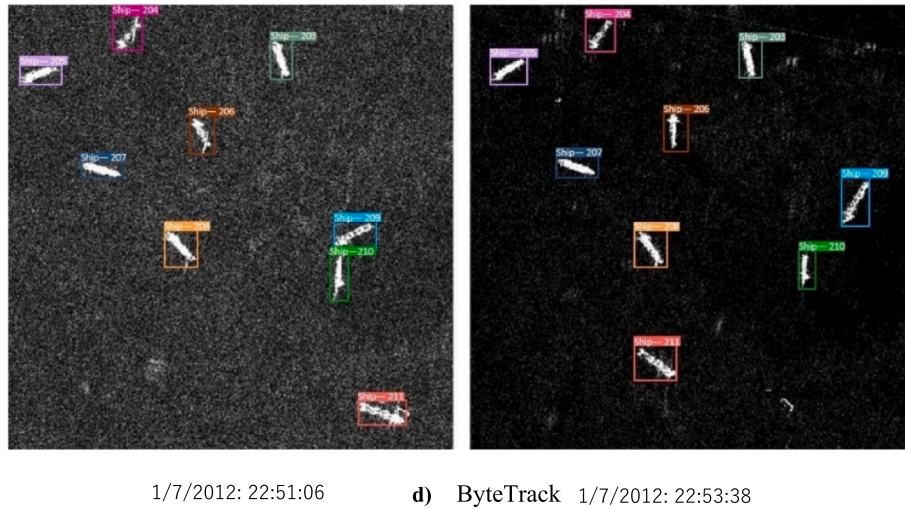


Fig. 10. (continued).

Table 6

Listed the results of various trackers techniques.

Model	MOTA (%)	HOTA (%)	MOTP (%)	IDF1 (%)	IDS	FPS/ (f/s)
FairMOT (Zhang et al., 2021b)	68.3	64.6	81.7	73.4	246	55.3
JDE (Wang et al., 2020)	65.3	61.5	83.9	68.7	303	56.6
MOTDT (Chen et al., 2018)	62.1	59.8	81.2	68.3	164	40.4
Sort (Bewley et al., 2016)	68.6	63.3	83.4	70.1	248	56.6
DeepSort (Wojke et al., 2017)	64.5	60.5	82.4	69.4	151	41.7
ByteTrack (Zhang et al., 2022)	70.3	64.2	83.9	75.0	136	56.7
Our model (YOLOv8-BYTE)	76.6	68.6	84.0	78.8	85	47.0

algorithm. While each component contributes uniquely to the model's effectiveness, their combined use leads to optimal results and balances accuracy and computational efficiency. This finding is crucial for further developments in multi-object tracking with SAR imagery, where precision and resource management are paramount.

The above optimization algorithms were tested on ship image instances in conjunction with the original YOLOv8 model, as shows in Fig. 6. Fig. 6 illustrated that the YOLOv8 model does not recognize some ships because of tiny targets created by occlusions between ships. In contrast, our improved model successfully identifies all ships. This success is due to our improved method, which skillfully balances attention between tiny objects addressed by shallow structures and large objects managed by deep structures, successfully decreasing miss detection and incorrect detection in scenarios with small targets or occlusions.

3.3. Analysis of multi object tracking of ship

3.3.1. Evaluation effects enhancement of model

To evaluate the efficiency of the multi-object technique, an experiment was conducted with four scenes. The results of the tracking experiments for these four scenes are presented in Table 4. In addition, the algorithm's efficiency was evaluated for these four scenes both before and after enhancement. The outcomes are listed in Table 5.

From Table 4, when comparing the efficiency of the different images,

it is evident that the tracking indexes for image 3 are comparatively worse than those for the other images. The tracking effect is optimal in image 1, where the ships are sparsely distributed, and occlusion interference is minimal. Table 5 shows a slight decrease in tracking speed when the improved YOLOv8 technique is utilized as a detector in contrast to the original YOLOv8 technique. However, all other accuracy indicators showed a significant improvement, confirming the efficiency of our proposed model. After implementing the refined ByteTrack technique in the current study, remarkable accuracy improvements were observed in HOTA, MOTA, and MOTP. This emphasizes that tracking efficiency can be successfully enhanced by directly predicting the tracking rectangle height and width information.

The developed algorithm shows a remarkable improvement in tracking performance, with an average HOTA of 68.6 %. This represents a significant enhancement of 4.8 % compared to the base algorithm and underlines the superior overall efficiency of the trajectory tracking technique presented in the current work. Furthermore, MOTA and MOTP show 7.5 % and 2.7 % improvements, respectively, in contrast to the base model. This indicates that the proposed model achieves higher accuracy and minimizes position errors in target tracking more effectively. The observed 6.8 % increase in IDF1 shows that our proposed model adapts more effectively and has higher reliability in target tracking. Notably, our improved algorithm has an ID value of 85, which is a 37.5 % reduction in contrast to the base model. Although the average frame rate of our model is slightly lower at 47 frames per second (FPS), a decrease of 9.4 frames per second opposed with the base algorithm, this trade-off results in significantly improved tracking accuracy and reliability. In summary, the proposed model in the current work significantly increases the accuracy of tracking and reliability compared to the base algorithm and achieves remarkable accuracy and speed in tracking vessel identifiers. This progress provides valuable technical support for the accurately management of vessel tracking.

The enhanced tracking outcomes of the ByteTrack algorithm are illustrated in Fig. 7. In this figure, the red rectangle indicates the tracking performance of the base ByteTrack technique, whereas the yellow rectangle denotes the tracking outcome obtained with the improved ByteTrack algorithm. Fig. 8 shows the proposed model's comparative performance and the base model's results. Fig. 8a shows that the original model cannot track the ship in some cases, as shown by the white circles in the figure. In contrast, Fig. 8b illustrations the efficiency of proposed model, which accurately tracks each vessel and significantly reduces the number of cases where it is not detected. Fig. 8a shows an anomaly in which false detection occurs. However, the proposed algorithm performs better because it has no false detections and

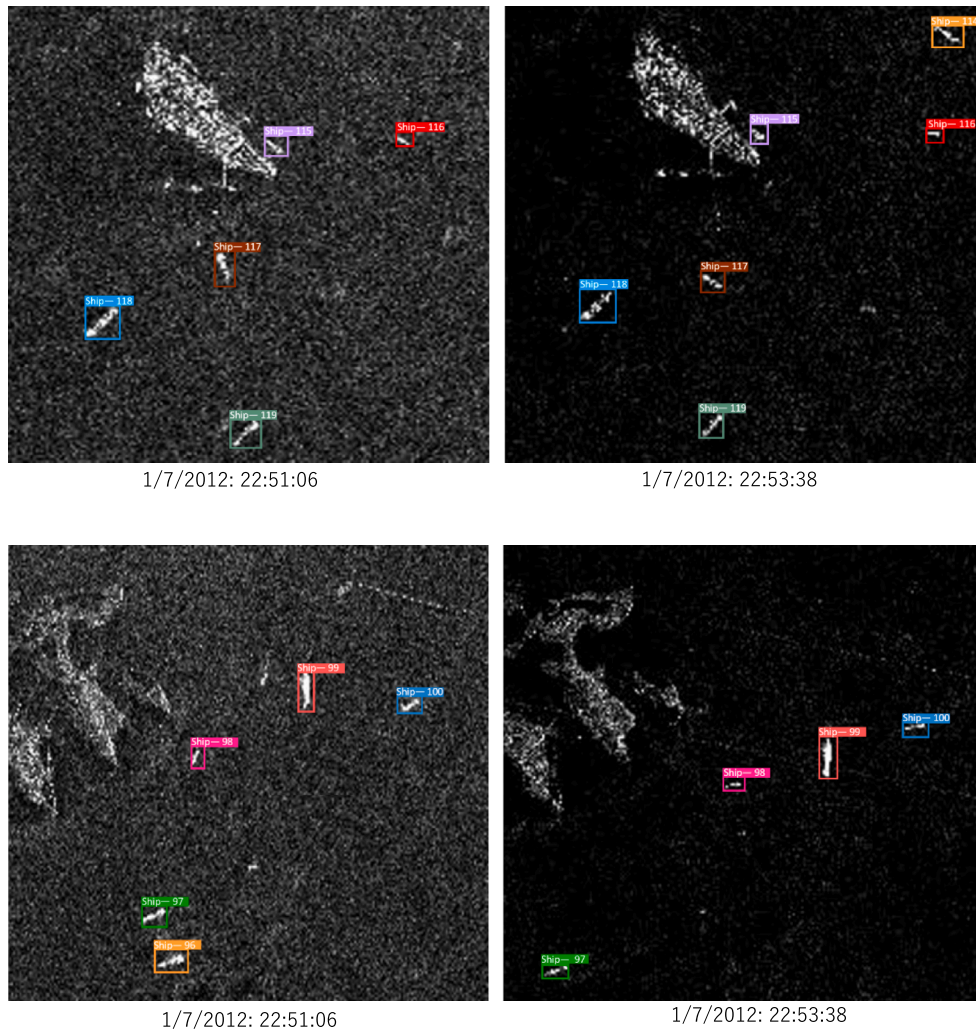


Fig 11. Ship tracking results in complex environment where non-ship targets are present.

improved detection and tracking capabilities.

3.3.2. Comparison of different tracking techniques

The vessel multi-object tracking dataset is used to evaluate the effectiveness of the MOT technique in this research. The results contrast with six other state-of-the-art object tracking models. Fig. 9 shows that our proposed tracking method has the largest HOTA measure, demonstrating its superior tracking capability. (See Fig. 10).

Table 6 shows that the tracking algorithm uses different MOT techniques. The accuracy of our proposed model is significantly higher, indicating its suitability as a detector for the MOT algorithms. When comparing these trackers techniques such as FairMOT (Zhang et al., 2021b), JDE (Wang et al., 2020), MOTDT (Chen et al., 2018), Sort (Bewley et al., 2016), DeepSort (Wojke et al., 2017), ByteTrack (Zhang et al., 2022), among these trackers MOTDT (Chen et al., 2018), and DeepSort (Wojke et al., 2017) performs the worst results. Although DeepSort adds associations to appearance features to cascade matching compared to Sort, extracting associations to appearance features has a lower computational cost, which ensures faster inference speed. However, this approach fails to distinguish objects with high similarity in appearance effectively. In the comparison between the Sort and ByteTrack algorithms, there is a slight decrease in frames per second (FPS) (1 ~ 2 FPS) as the complexity of the model rise, but there is a significant improvement in tracking accuracy. Our proposed YOLOv8-BYTE model outperforms others in the accuracy indicator, including MOTA, HOTA, MOTP, and IDF1. The proposed YOLOv8-BYTE method enhances the

efficiency of object tracking by eliminating noise and background information from the low-score recognition outcomes while effectively utilizing the high-score and low-score recognition patches. The proposed model records only 85 ID switches, significantly fewer than the six other models. This indicates that our model experiences fewer ID switches and performs well when tracking multiple objects in different scenes. The results indicate the suitability of the proposed method for tracking vessels with multiple objects in different scenarios. Our method exhibits robust detection and tracking capabilities despite the lack of missed tracking targets.

As for false recognition, DeepSort, Sort and eliminates low-scoring recognition rectangles, preventing false detection cases. However, the ByteTrack technique retains a low-scoring recognition rectangle when similar objects appear and can thus cause false detection cases. Nevertheless, our proposed model enhances the YOLOv8 model and incorporates it with improved ByteTrack, resulting in stable tracking effectiveness in different scenarios. To summarize, our proposed algorithm outperforms among these trackers MOT algorithms. It shows robust tracking effectiveness in different scenarios and provides valuable technical support for precise vessel management under various conditions.

3.4. Ship tracking in complex environment

To provide further clarity on how ship tracking can be handled under difficult conditions, especially when non ship targets are available, we

performed additional evaluations of our proposed multi-target tracking algorithm. Fig. 11, our YOLOv8-BYTE algorithm shows robust performance by accurately tracking ships in SAR short time sequence images without missing ship tracking. Most importantly, YOLOv8-BYTE maintains its exceptional tracking performance even in complex environments with vessels of varying sizes and the presence of other targets that are not vessels. This resilience emphasizes the efficiency of our algorithm in real-world scenarios where diverse and cluttered backgrounds can pose a challenge. We believe that these results further validate the practicality and reliability of our proposed approach and increase its relevance for the advancement of ship tracking technology in maritime domains.

4. Conclusion

In this study, we present a novel ship tracking algorithm called YOLOv8-BYTE, which is specifically tailored for disaster response leveraging GeoAI. Recognizing the critical role of vessel detection and tracking in emergency rescue scenarios, our algorithm is designed to quickly and accurately determine the position and status of vessels to enable efficient deployment of rescue teams in disaster areas. While vessel tracking technology has already been extensively studied for optical image dataset, our motivation for SAR image dataset addresses the unique challenges posed by defocus interference, a high false alarm rate and a lack of prior samples. By optimizing the YOLOv8 detection network using a diffusion model-based training method for data augmentation, we improve the robustness of the algorithm to scale, rotational and translational deformations. The introduction of an enhanced swin transformer as a feature extraction network helps to improve the visualization capability of the detection network and ensure effective tracking of multiple ships in different image scenes. In addition, the refinement of the ByteTrack algorithm with a novel state vector improves the accuracy of the tracking box. The proposed MOT algorithm demonstrates SOTA tracking performance on short-time sequence SAR imagery and provides a compelling solution for ship tracking in disaster scenarios. This research contributes to the development of ship tracking technologies for SAR imagery and significantly improves the overall resilience of maritime transportation systems during emergency response operations.

CRedit authorship contribution statement

Muhammad Yasir: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Liu Shanwei:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition. **Xu Mingming:** Writing – review & editing, Methodology. **Wan Jianhua:** Writing – review & editing, Software, Methodology, Data curation, Conceptualization. **Sheng Hui:** Writing – review & editing, Methodology. **Shah Nazir:** Writing – review & editing, Visualization. **Xin Zhang:** Writing – review & editing, Visualization. **Arife Tugsan Isiacik Colak:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

National Key Research and Development Program of China

(2017YFC1405600).

References

- Bewley, A., Ge, Z., Ott, L., Ramos, F., Upcroft, B., 2016. Simple online and realtime tracking, 2016 IEEE international conference on image processing (ICIP). IEEE, pp. 3464–3468.
- Bhattacharjee, S., Shanmugam, P., Das, S., 2023. A Seep-Learning-Based Lightweight Model for Ship Localizations in SAR Images. IEEE Access.
- Chen, L., Cai, X., Xing, J., Li, Z., Zhu, W., Yuan, Z., Fang, Z., 2023. Towards transparent deep learning for surface water detection from SAR imagery. *Int. J. Appl. Earth Obs. Geoinf.* 118, 103287.
- Chen, L., Ai, H., Zhuang, Z., Shang, C., 2018. Real-time multiple people tracking with deeply learned candidate selection and person re-identification, 2018 IEEE international conference on multimedia and expo (ICME). IEEE, pp. 1–6.
- Chen, L., Weng, T., Xing, J., Li, Z., Yuan, Z., Pan, Z., Tan, S., Luo, R., 2021. Employing deep learning for automatic river bridge detection from SAR images based on adaptively effective feature fusion. *Int. J. Appl. Earth Obs. Geoinf.* 102, 102425.
- Cheng, S., Zhu, Y., Wu, S., 2023. Deep learning based efficient ship detection from drone-captured images for maritime surveillance. *Ocean Eng.* 285, 115440.
- Cong-An, X., Hang, S., Long, G., Jun-Feng, W., Wen-Jun, Y., Tao, J., Hai-Yang, W., 2022. Feature aligned ship detection based on improved RPDet in SAR images. *Displays* 74, 102191.
- Ding, K., Yang, J., Lin, H., Wang, Z., Wang, D., Wang, X., Ni, K., Zhou, Q., 2023. Towards real-time detection of ships and wakes with lightweight deep learning model in Gaofen-3 SAR images. *Remote Sens. Environ.* 284, 113345.
- Escoria-Gutierrez, J., Gamarra, M., Beñito, K., Soto, C., Mansour, R.F., 2022. Intelligent deep learning-enabled autonomous small ship detection and classification model. *Comput. Electr. Eng.* 100, 107871.
- Ghiasi, G., Cui, Y., Srinivas, A., Qian, R., Lin, T.-Y., Cubuk, E.D., Le, Q.V., Zoph, B., 2021. Simple copy-paste is a strong data augmentation method for instance segmentation, *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2918–2928.
- Humayun, M.F., Bhatti, F.A., Khurshid, K., 2023. iVision MRSSD: A comprehensive multi-resolution SAR ship detection dataset for state of the art satellite based maritime surveillance applications. *Data Brief* 50, 109505.
- Jiang, B., Ma, X., Lu, Y., Li, Y., Feng, L., Shi, Z., 2019. Ship detection in spaceborne infrared images based on convolutional neural networks and synthetic targets. *Infrared Phys. Technol.* 97, 229–234.
- Kong, W., Liu, S., Xu, M., Yasir, M., Wang, D., Liu, W., 2023. Lightweight algorithm for multi-scale ship detection based on high-resolution SAR images. *Int. J. Remote Sens.* 44, 1390–1415.
- Li, W., Li, F., Li, Z., 2022b. CMFTNet: Multiple fish tracking based on counterpoised JointNet. *Comput. Electron. Agric.* 198, 107018.
- Li, J., Xu, C., Su, H., Gao, L., Wang, T., 2022a. Deep learning for SAR ship detection: Past, present and future. *Remote Sens. (Basel)* 14, 2712.
- Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021. Swin transformer: Hierarchical vision transformer using shifted windows, *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 10012–10022.
- Liu, S., Kong, W., Chen, X., Xu, M., Yasir, M., Zhao, L., Li, J., 2022. Multi-scale ship detection algorithm based on a lightweight neural network for spaceborne SAR images. *Remote Sens. (Basel)* 14, 1149.
- Liu, X., Zhang, Y., Zou, H., Wang, F., Cheng, X., Wu, W., Liu, X., Li, Y., 2023. Multi-source knowledge graph reasoning for ocean oil spill detection from satellite SAR images. *Int. J. Appl. Earth Obs. Geoinf.* 116, 103153.
- Lou, X., Liu, Y., Xiong, Z., Wang, H., 2022. Generative knowledge transfer for ship detection in SAR images. *Comput. Electr. Eng.* 101, 108041.
- Mao, W., Wang, X., Liu, G., Pirasteh, S., Zhang, R., Lin, H., Xie, Y., Xiang, W., Ma, Z., Ma, P., 2023. Time series insar ionospheric delay estimation, correction, and ground deformation monitoring with reformulating range split-spectrum interferometry. *IEEE Trans. Geosci. Remote Sens.*
- Matasci, G., Plante, J., Kasa, K., Mousavi, P., Stewart, A., Macdonald, A., Webster, A., Busler, J., 2021. Deep learning for vessel detection and identification from spaceborne optical imagery. *ISPRS Ann. Photogram. Remote Sens. Spatial Inform. Sci.* 3, 303–310.
- Owda, A., Dall, J., Badger, M., Cavar, D., 2023. Improving SAR wind retrieval through automatic anomalous pixel detection. *Int. J. Appl. Earth Obs. Geoinf.* 122, 103444.
- Peng, J., Wang, C., Wan, F., Wu, Y., Wang, Y., Tai, Y., Wang, C., Li, J., Huang, F., Fu, Y., 2020. Chained-tracker: Chaining paired attentive regression results for end-to-end joint multiple-object detection and tracking, *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IV* 16. Springer, pp. 145–161.
- Tan, C., Li, C., He, D., Song, H., 2022. Towards real-time tracking and counting of seedlings with a one-stage detector and optical flow. *Comput. Electron. Agric.* 193, 106683.
- Wang, M., Larsen, M.L., Liu, D., Winters, J.F., Rault, J.-L., Norton, T., 2022. Towards re-identification for long-term tracking of group housed pigs. *Biosyst. Eng.* 222, 71–81.
- Wang, Z., Zheng, L., Liu, Y., Li, Y., Wang, S., 2020. Towards real-time multi-object tracking. *Eur. Conf. Comput. Vis. Springer* 107–122.
- Wojke, N., Bewley, A., Paulus, D., 2017. Simple online and realtime tracking with a deep association metric, 2017 IEEE international conference on image processing (ICIP). IEEE, pp. 3645–3649.
- Xia, G.-S., Bai, X., Ding, J., Zhu, Z., Belongie, S., Luo, J., Datcu, M., Pelillo, M., Zhang, L., 2018. DOTA: A large-scale dataset for object detection in aerial images, *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3974–3983.

- Yang, S., Wang, W., Gao, S., Deng, Z., 2023. Strawberry ripeness detection based on YOLOv8 algorithm fused with LW-swin transformer. *Comput. Electron. Agric.* 215, 108360.
- Yasir, M., Jianhua, W., Mingming, X., Hui, S., Zhe, Z., Shanwei, L., Colak, A.T.I., Hossain, M.S., 2023a. Ship detection based on deep learning using SAR imagery: A systematic literature review. *Soft. Comput.* 27, 63–84.
- Yasir, M., Shanwei, L., Mingming, X., Hui, S., Hossain, M.S., Colak, A.T.I., Wang, D., Jianhua, W., Dang, K.B., 2023b. Multi-scale ship target detection using SAR images based on improved Yolov5. *Front. Mar. Sci.* 9, 1086140.
- Yasir, M., Zhan, L., Liu, S., Wan, J., Hossain, M.S., Isiacik Colak, A.T., Liu, M., Islam, Q. U., Raza Mehdi, S., Yang, Q., 2023c. Instance segmentation ship detection based on improved Yolov7 using complex background SAR images. *Front. Mar. Sci.* 10, 1113669.
- Yasir, M., Liu, S., Mingming, X., Wan, J., Pirasteh, S., Dang, K.B., 2024. ShipGeoNet: SAR image-based geometric feature extraction of ships using convolutional neural networks. *IEEE Trans. Geosci. Remote Sens.*
- Zha, C., Min, W., Han, Q., Xiong, X., Wang, Q., Xiang, H., 2023. SAR ship detection based on salience region extraction and multi-branch attention. *Int. J. Appl. Earth Obs. Geoinf.* 123, 103489.
- Zhang, X., Li, Y., Li, F., Jiang, H., Wang, Y., Zhang, L., Zheng, L., Ding, Z., 2024. Ship-go: SAR ship images inpainting via instance-to-image generative diffusion models. *ISPRS J. Photogramm. Remote Sens.* 207, 203–217.
- Zhang, Y., Wang, C., Wang, X., Zeng, W., Liu, W., 2021b. Fairmot: On the fairness of detection and re-identification in multiple object tracking. *Int. J. Comput. Vis.* 129, 3069–3087.
- Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., Wang, X., 2022. Bytetrack: Multi-object tracking by associating every detection box. *Eur. Conf. Comput. Vis. Springer* 1–21.
- Zhang, T., Zhang, X., Liu, C., Shi, J., Wei, S., Ahmad, I., Zhan, X., Zhou, Y., Pan, D., Li, J., 2021a. Balance learning for ship detection from synthetic aperture radar remote sensing imagery. *ISPRS J. Photogramm. Remote Sens.* 182, 190–207.
- Zheng, Z., Li, J., Qin, L., 2023. YOLO-BYTE: An efficient multi-object tracking algorithm for automatic monitoring of dairy cows. *Comput. Electron. Agric.* 209, 107857.
- Zhou, X., Koltun, V., Krähenbühl, P., 2020. Tracking objects as points. *Eur. Conf. Comput. Vis. Springer* 474–490.